Table 1: When does a learned KKL observer win (C3)? sequence OOD, dev plants: number of plants on which HyperKKL ($r = 4$, the single learned method) beats the comparator (paired 95% CI excludes 0). Per-plant numbers: Table **??**.

		(a) vs model-based filters					(b) vs data-driven	
scenario	model err.	best filter	working	nominal	rob./orac.	failed	best data-driven	NLOX
exact	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	pend. 0/30	4/6
exact+outliers	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	15/48	pend. 0/30	1/6 (4L)
colored_t	0.00	0/6 (6L)	0/6 (6L)	0/6 (6L)	0/6 (6L)	1/48	0/6 (6L)	5/6
mismatch10	0.09	4/6 (1L)	4/6 (1L)	4/6 (1L)	4/6 (1L)	13/48	0/6 (5L)	4/6
mismatch20	0.17	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	15/48	1/6 (5L)	5/6 (1L)
drift30 (post-train)	0.24	0/6 (5L)	0/5 (5L)*	2/6 (3L)	1/6 (5L)	15/48	pend. 0/30	3/6 (1L)
mismatch30	0.24	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	16/48	0/6 (4L)	4/6
mismatch50	0.37	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	16/48	0/6 (4L)	4/6
partial	0.44	5/6 (1L)	4/5 (1L)*	5/6 (1L)	5/6 (1L)	14/48	1/6 (4L)	4/6 (1L)
realistic	0.63	6/6	4/4*	6/6	6/6	24/48	2/6 (4L)	5/6 (1L)

(kL): the comparator is significantly better on k plants. (a) Model-based filters: EKF, UKF, PF, KKL-MB and their Huber and oracle variants. A filter is FAILED when its NRMSE exceeds climatology (the constant mean-state predictor); 'working' = the best filter that did not fail (*: no filter works on some plants). (b) Data-driven baselines trained on the same real-plant data: NLOX, Koopman KF (incl. bilinear), grey-box EKF/UKF/EKF-oracle; 'pend. m/N' = merged in m of N runs. Model error: median over dev plants.